[redacted]

TEST REPORT

Part No.	:	SG-50/304
Rated Power	:	50kVA
Series No.	:	B22120478
Finish Date	:	2022/12/31

Check : [redacted]

Approve : [redacted]

[redacted]

[redacted]

Tel: +86-757-86256888 FAX: +86-757-8625-6886

[redacted]

TEST REPORT

Manuf. No. B22120478

1. Technical Specification.

1) Part Name:	Dry-type Transformer
2) Series No.:	B22120478
3) P/N:	SG-50/304
4) Rated Power:	50kVA
5) Primary Rated Current:	137.5A
6) Secondary Rated Current:	65.6-68.7A
7) Primary Rated Voltage:	210V
8) Secondary Rated Voltage:	440-420V
9) Phase:	Three Phase
10) Rated Frequency:	50/60Hz
11) Vector Group:	Dy11
12) Cooling Method:	AN
13) Manufacture Date:	2022/12/31
14) Insulation Grade:	H
15) Applicable Condition:	Outdoor
16) Total Weight:	400kg

2. Standard of Test.

JEC-2200-2014

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3. Visual Inspection.

Items	Judgment standard	Conclusion
Dimension	To confirm dimensions are in accordance with the approved drawing.	OK
Code of terminal	To confirm dimensions are in accordance with the approved drawing.	OK
Tightening check	To ensure that the specified tightening torque.	OK
Tested by	[redacted]	Date 2022/12/31

4. DC Resistance Test. Temperature: 16.0℃ Humidity: 54.0%

Primary Winding(mΩ)

Test position	Design value	uv	vw	wu	Maximum deviation(±5%)
210V	8.558	8.830	8.810	8.744	3.18

Secondary Winding(mΩ)

Test position	Design value	UV	VW	WU	Maximum deviation(±5%)
440V	68.97	66.30	66.39	66.39	-3.87
420V	65.83	63.18	63.29	63.24	-4.03
Conclusion	OK	Tested by	[redacted]	Date	2022/12/31

5.Insulation Resistance Test. Temperature: 16.0℃ Humidity: 54.0%

Test position	Test results (MΩ)	Judgment standard(MΩ)	Conclusion	Test equipment
Primary----secondary & Earth	2500	≥ 100	OK	2500V/2500MΩ
Secondary---primary & Earth	2500	≥ 100	OK	
Tested by	[redacted]	Date	2022/12/31	

6. Ratio Group Test. Temperature: 16.0℃ Humidity: 54.0%

Test position	Connection	Standard ratio	uv/UV	vw/VW	wu/WU	Maximum deviation	Test connection
210/440	Dy11	0.4773	0.4770	0.4763	0.4767	-0.19	Dy11
210/420		0.5000	0.4986	0.4980	0.4982	-0.39	
Conclusion		OK	Tested by	[redacted]	Date	2022/12/31	

7. Applied Voltage Withstand Test. Temperature: 16.0℃ Humidity: 54.0%

(Judgment standard:the power is not tripping operation suddenly,not obvious discharge and without sparks sounds)

Test position	Test voltage(kV)	Current(mA)	During time(s)	Conclusion
Primary----secondary & Earth	2.0	4.3	60	OK
Secondary---primary & Earth	4.0	1.7	60	OK
Tested by	[redacted]	Date	2022/12/31	

[redacted]

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8.Induced Voltage Withstand Test Temperature: 16.0℃ Humidity: 54.0%

(Judgment standard:the power is not tripping operation suddenly,not obvious discharge and without sparks sounds)

Test position(V)	Test voltage(V)	Frequency (Hz)	During time(s)	Conclusion
210	420	150	48	OK
Tested by	[redacted]	Date	2022/12/31	

9.Short-circuit Impedance and Load Loss. (60Hz) Temperature: 16.0℃ Humidity: 54.0%

Test position	Test voltage (V)	Current (A)	Test loss (W)	%IZ	140℃				
					R%	X%	ε%	%IZ	Load loss
440V	19.66	65.6	703	4.47	2.09	4.24	2.18	4.73	1043
420V	18.89	68.7	729	4.50	2.15	4.25	2.15	4.77	1077
Conclusion	OK	Tested by	[redacted]	Date	2022/12/31				

10. Test on No-load Current and No-load Loss.(60Hz) Temperature: 16.0℃ Humidity: 54.0%

Test voltage(V)	No-load current(A)	No-load loss (214W)	No-load current (≤3%)	Conclusion
210	0.99	161	0.72	OK
Tested by	[redacted]	Date	2022/12/31	

11. Efficiency 140℃ Judgment standard: ≥97.40 %

$$\eta = \frac{Se}{Se + P_{NoloadLoss} + P_{Full\text{-}LoadLoss}} = 97.58\ \%$$

Conclusion: OK

Tested by [redacted] Date 2022/12/31